

CLIENT SUMMARY REPORT

Customer Shopping Behavior

Understanding what drives customer decisions, loyalty and repeat purchases

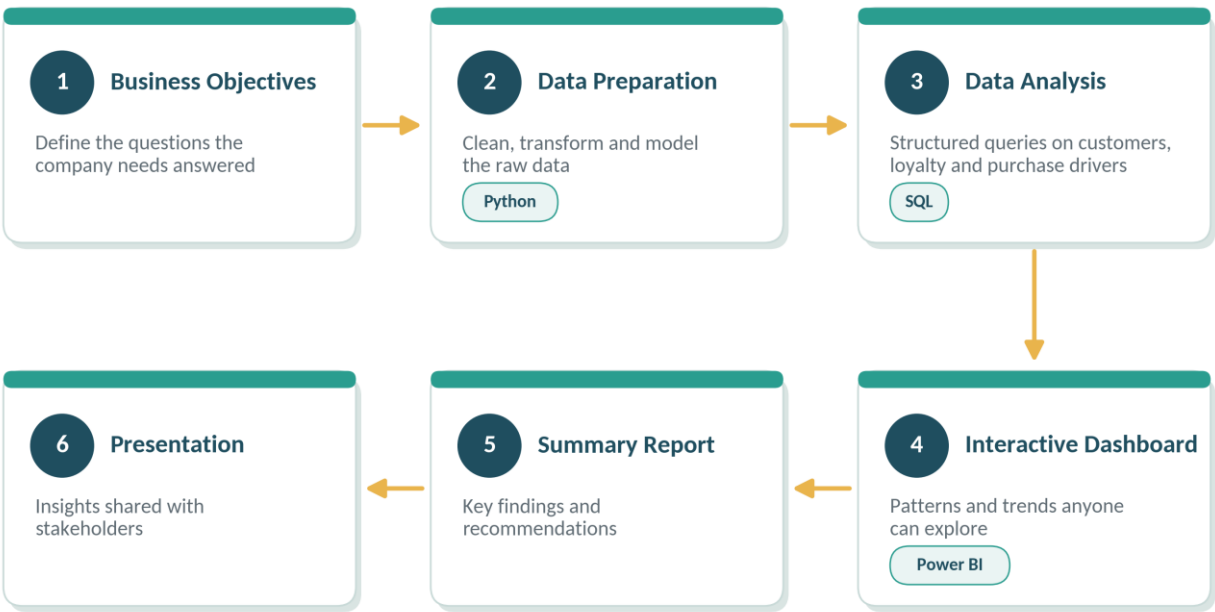
1. Introduction

Thank you for the opportunity to explore your customer shopping data. This report summarizes what we did, what we found, and the steps we recommend to help you grow sales, strengthen customer satisfaction and build long-term loyalty.

We analyzed 3,900 purchases to understand how customers shop across different demographics, product categories and shipping options. The sections that follow walk you through our approach, the analysis, and our recommendations. An interactive Power BI dashboard accompanies this report so your team can keep exploring the results.

2. Our Approach

Our work followed six connected steps, taking your data from raw records to clear, actionable insights:



3. What We Delivered

The project brings together four deliverables, each building on the one before it:

Deliverable	Tool	What It Covers
Data Preparation & Modeling	Python	Cleaning and transforming the raw dataset so it is ready for analysis.
Data Analysis	SQL	Organizing the data in a structured format, simulating business transactions, and running queries on customer segments, loyalty and purchase drivers.
Visualization & Insights	Power BI	An interactive dashboard that highlights key patterns and trends and supports data-driven decisions.
Report & Presentation	Report / Slides	A clear project report with key findings and recommendations, plus a presentation that communicates them visually to stakeholders.

4. Analysis Questions

The analysis is designed to answer these ten questions:

1. What is the total revenue generated by male vs. female customers?
2. Which customers use a discount but still spend more than the average purchase amount?
3. Which are the top 5 products with the highest average review rating?
4. How does the average purchase amount compare between standard and express shipping?
5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.
6. Which 5 products have the highest percentage of purchases with discounts applied?
7. Segment customers into New, Returning and Loyal based on their number of previous purchases, and show the count of each segment.
8. What are the top 3 most purchased products within each category?
9. Are repeat buyers (more than 5 previous purchases) also likely to subscribe?
10. What is the revenue distribution across each age group?

5. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across a range of product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences and subscription behavior, so the business can make smarter strategic decisions.

Dataset Summary

3,900 Rows	18 Columns	37 Missing values (Review Rating)
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The dataset captures three main kinds of information:

Feature Group	What It Includes
Customer Demographics	Age, gender, location and subscription status.
Purchase Details	Item purchased, category, purchase amount, season, size and color.
Shopping Behavior	Discount applied, promo code used, previous purchases, purchase frequency, review rating and shipping type.

6. Exploratory Data Analysis (Python)

Before any analysis, we cleaned and prepared the data in Python so that every number in this report can be trusted. The snippets below show exactly what we ran and what the data looked like afterward.

Measure	Before	After
Rows	3,900	3,900
Columns	18	19
Missing Review Ratings	37	0
Column Names	Mixed case, with spaces	Lower snake case
Duplicate Columns	1	0

Step 1: Load and Inspect the Data

We loaded the file into Pandas and checked its structure. The output confirms 3,900 rows and 18 columns, and shows that Review Rating holds only 3,863 values, which means 37 are missing.

PYTHON CODE

```
import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')
df.info()
```

OUTPUT

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Customer ID           3900 non-null   int64
1   Age                   3900 non-null   int64
2   Gender                3900 non-null   object
```

```

3 Item Purchased      3900 non-null object
4 Category            3900 non-null object
5 Purchase Amount (USD) 3900 non-null int64
6 Location            3900 non-null object
7 Size                3900 non-null object
8 Color               3900 non-null object
9 Season              3900 non-null object
10 Review Rating       3863 non-null float64
11 Subscription Status 3900 non-null object
12 Shipping Type       3900 non-null object
13 Discount Applied    3900 non-null object
14 Promo Code Used     3900 non-null object
15 Previous Purchases  3900 non-null int64
16 Payment Method      3900 non-null object
17 Frequency of Purchases 3900 non-null object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB

```

Step 2: Find and Fill Missing Values

Rather than using one overall median, we filled each missing rating with the median rating of its own product category. The check before and after shows the 37 gaps are gone.

PYTHON CODE

```

# Fill missing ratings with the median rating of each product category
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(
    lambda x: x.fillna(x.median())
)

```

BEFORE

```
df.isnull().sum()
```

```

Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD) 0
Location         0
Size             0
Color            0
Season           0
Review Rating    37
Subscription Status 0
Shipping Type    0
Discount Applied 0
Promo Code Used  0
Previous Purchases 0
Payment Method   0
Frequency of Purchases 0
dtype: int64

```

AFTER

```
df.isnull().sum()
```

```

Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD) 0
Location         0
Size             0
Color            0
Season           0
Review Rating    0
Subscription Status 0
Shipping Type    0
Discount Applied 0
Promo Code Used  0
Previous Purchases 0
Payment Method   0
Frequency of Purchases 0
dtype: int64

```

GOOD TO KNOW

Missing ratings were replaced with the median for their own category, not the overall median. This keeps each product group's ratings more realistic.

Step 3: Standardize the Column Names

Mixed upper and lower case and spaces make analysis in Python and SQL error-prone, so we converted every column name to lower snake case.

PYTHON CODE

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

BEFORE

Customer ID
Age
Gender
Item Purchased
Category
Purchase Amount (USD)
Location
Size
Color
Season
Review Rating
Subscription Status
Shipping Type
Discount Applied
Promo Code Used
Previous Purchases
Payment Method
Frequency of Purchases

AFTER

customer_id
age
gender
item_purchased
category
purchase_amount
location
size
color
season
review_rating
subscription_status
shipping_type
discount_applied
promo_code_used
previous_purchases
payment_method
frequency_of_purchases

Step 4: Create New Columns

Age Groups

We split customer ages into four quartile-based groups, so each group holds roughly a quarter of the customers rather than a fixed age range.

PYTHON CODE

```
labels = ['Young Adults', 'Adults', 'Middle-aged', 'Seniors']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)

df[['age', 'age_group']].head(10)
```

OUTPUT

```
   age  age_group
0   55  Middle-aged
1   19  Young Adults
2   50  Middle-aged
3   21  Young Adults
4   45  Middle-aged
5   46  Middle-aged
6   63    Seniors
7   27  Young Adults
8   26  Young Adults
9   57  Middle-aged
```

Purchase Frequency in Days

We turned the text purchase frequency into a number of days, which is much easier to analyze.

PYTHON CODE

```
frequency_mapping = {
    'Fortnightly': 14, 'Weekly': 7, 'Monthly': 30, 'Quarterly': 90,
    'Bi-Weekly': 14, 'Annually': 365, 'Every 3 Months': 90
}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)

df[['purchase_frequency_days', 'frequency_of_purchases']].head(10)
```

OUTPUT

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

Step 5: Remove a Duplicate Column

The discount_applied and promo_code_used columns looked identical. A direct comparison confirmed they match on every row, so we kept discount_applied and dropped promo_code_used.

PYTHON CODE

```
# Are the two columns identical?
(df['discount_applied'] == df['promo_code_used']).all()
```

OUTPUT

```
True
```

PYTHON CODE

```
df = df.drop('promo_code_used', axis=1)
df.columns
```

OUTPUT

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'previous_purchases', 'payment_method',
      'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
      dtype='object')
```

Step 6: Check the Cleaned Data

Summary statistics on the cleaned table show an average purchase of about \$59.76 and an average review rating of 3.75, the same headline figures that appear on the dashboard.

PYTHON CODE

```
df.describe()
```

OUTPUT

	customer_id	age	purchase_amount	review_rating \
count	3900.000000	3900.000000	3900.000000	3900.000000

mean	1950.500000	44.068462	59.764359	3.750051
std	1125.977353	15.207589	23.685392	0.713590
min	1.000000	18.000000	20.000000	2.500000
25%	975.750000	31.000000	39.000000	3.100000
50%	1950.500000	44.000000	60.000000	3.800000
75%	2925.250000	57.000000	81.000000	4.400000
max	3900.000000	70.000000	100.000000	5.000000

	previous_purchases	purchase_frequency_days
count	3900.000000	3900.000000
mean	25.351538	89.133077
std	14.447125	119.037566
min	1.000000	7.000000
25%	13.000000	14.000000
50%	25.000000	30.000000
75%	38.000000	90.000000
max	50.000000	365.000000

Step 7: Load the Data Into MySQL

Finally, we loaded the cleaned table into MySQL for the SQL analysis. The connection asks for the password at the prompt, so it is never stored in the code. All 3,900 rows loaded successfully.

PYTHON CODE

```
from getpass import getpass
from sqlalchemy import create_engine
from sqlalchemy.engine import URL

url = URL.create(
    "mysql+pymysql",
    username="root",
    password=getpass("MySQL password: "),
    host="localhost",
    port=3306,
    database="customer_behavior",
)
engine = create_engine(url)

df.to_sql("customer_shopping_behavior", con=engine, if_exists="replace", index=False)
```

ROWS LOADED

3900

7. Data Analysis (SQL)

Using MySQL Workbench, we ran structured queries to answer the key business questions. Each result is shown below its question.

Revenue by Gender

Total revenue generated by male vs. female customers.

	gender	revenue
▶	Male	157890
	Female	75191

High-Spending Discount Users

Customers who used a discount but still spent above the average purchase amount.

	customer_id	purchase_amount
▶	1480	100
	1413	100
	96	100
	1422	100
	1457	100
	1209	100
	194	100
	770	100
	1406	100
	43	100

Average Review Rating

The top 5 products with the highest average review rating.

	item_purchased	Average_Product_rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.78

Shipping Type Comparison

Average purchase amount for standard vs. express shipping.

	shipping_type	average_purchase_amount	total_purchases
▶	Express	60.48	39067
	Standard	58.46	38233

Spending vs. Subscription Status

Whether subscribed customers spend more than non-subscribers.

	no_of_customers	subscription_status	average_spent	total_revenue
▶	1053	Yes	59.49	62645
	2847	No	59.87	170436

Discounted Purchases

The 5 products with the highest percentage of purchases made with a discount.

	item_purchased	discount_rate
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

Customer Segments

New, Returning and Loyal customers, based on number of previous purchases.

	customer_segment	Number_of_Customers
▶	Loyal	3116
	Returning	701
	New	83

Top 3 Products per Category

The most purchased products within each category.

	item_rank	category	item_purchased	total_orders
▶	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

Repeat Buyers & Subscriptions

Whether customers with more than 5 purchases are more likely to subscribe.

	subscription_status	repeat_buyers
▶	Yes	958
	No	2518

Revenue by Age Group

Total revenue distribution across each age group.

	age_group	total_revenue
▶	Young Adults	62143
	Middle-aged	59197
	Adults	55978
	Seniors	55763

8. Business Recommendations

Based on the analysis, these are the actions we recommend:

Recommendation	How to Put It into Action
Boost Subscriptions	Promote exclusive benefits for subscribers to encourage more sign-ups.
Customer Loyalty Programs	Reward repeat buyers to help move them into the Loyal segment.
Review Discount Policy	Balance the sales boost from discounts with healthy margin control.
Product Positioning	Feature top-rated products prominently in marketing campaigns.
Targeted Marketing	Focus efforts on high-revenue age groups and express-shipping customers.
Gender-Focused Strategy	Target male buyers, who bring in more revenue, or offer rewards and discounts to engage more female customers.

Thank you for reading. We hope these insights help turn shopping data into confident, customer-friendly decisions.